PAWAN AJAY PRASAD

AI Engineer | Generative AI & RAG Systems | Machine Learning

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PROFESSIONAL SUMMARY

Entry-level AI Engineer with hands-on experience building and deploying 3+ end-to-end GenAI, RAG, and machine learning projects using Python, LangChain, Gemini, ChromaDB, and FastAPI. Delivered an enterprise RAG knowledge assistant with 100% retrieval accuracy (Hit@1/3/5), an XGBoost predictive maintenance model at 98%+ accuracy, and a Power BI credit-risk dashboard analyzing 100,000+ records. Skilled in SQL, EDA, and statistical analysis, with practical experience across the full AI development lifecycle — data pipelines, model deployment, and REST API integration.

SKILLS

Generative AI & AI Engineering: GenAI, Retrieval-Augmented Generation (RAG), LangChain, Vector Databases (ChromaDB), Semantic Search, Embeddings, LLM Integration (Gemini), Prompt Engineering, REST API Development, FastAPI, AI Application Development, Model Deployment

Machine Learning: Scikit-learn, XGBoost, Feature Engineering, Predictive Modeling, Predictive Analytics, SMOTE, Hypothesis Testing, Model Evaluation, Model Deployment

Programming & Data: Python, SQL, PostgreSQL, Pandas, NumPy, Excel

Business Intelligence: Power BI, DAX, Tableau, Dashboard Development, KPIs, Data Reporting

Research & Analytics: Exploratory Data Analysis (EDA), Statistical Analysis, Data Interpretation, Trend Analysis, Data Visualization, Insight Generation

Platforms & Tools: GitHub, Microsoft Azure, Kubernetes, Streamlit, Docker

TRAINING EXPERIENCE

AlmaBetter — AI Engineering / Data Analyst Trainee — **Jan 2025 – Present**

- Implemented AI/GenAI workflows using Python, LangChain, Gemini, RAG, embeddings, and Streamlit for document-based Q&A.
- Built RAG pipelines covering document ingestion, chunking, semantic retrieval, and citation-grounded LLM responses, improving answer accuracy for internal knowledge search.
- Applied ML, EDA, SQL, and BI across 15+ datasets leveraging Scikit-learn/XGBoost, 100K+ data points, and 20+ KPIs to surface predictive and business trends.
- Automated data-cleaning scripts across 50+ SQL queries, reducing manual data-preparation time by roughly 30%.
- Communicated analytical findings to a cohort of 50+ peers and mentors to support business-ready decisions.

PROJECTS

Infosys AI Knowledge Assistant — Enterprise RAG Platform — **AlmaBetter Verified Project**

Python, FastAPI, LangChain, Gemini, ChromaDB, RAG | GitHub | Live Demo

- Architected and delivered an enterprise RAG knowledge assistant with LangChain, Gemini 3.6 Flash, and ChromaDB, serving fully-cited responses across 5 department domains with role-based access control (RBAC).
- Engineered PDF ingestion and semantic retrieval via PyMuPDF, 1K/200-token chunking, Gemini embeddings, and department/document metadata filtering — cutting document-lookup time versus manual search.
- Achieved 100% Hit@1/3/5, MRR 1.0, and RBAC pass rate across 25 retrieval test cases, with 5/5 generation evaluations passed.

AI-Based Predictive Maintenance & Failure Diagnosis — **AlmaBetter Verified Project**

Python, XGBoost, Scikit-learn, Streamlit, GenAI | GitHub | Live Demo

- Investigated industrial machine-failure data across 10,000+ sensor readings through EDA, hypothesis testing, and feature engineering, applying SMOTE to correct class imbalance and isolate key failure risk factors.
- Developed and deployed an XGBoost-based predictive maintenance model achieving 98%+ accuracy, paired with GenAI-based diagnostics on Streamlit — cutting failure-diagnosis effort by nearly 40% and shortening turnaround on repair recommendations.
- Presented model performance and risk findings, strengthening the team's data-driven planning process.

PaisaBazaar — Credit Risk Analysis & Dashboard — **AlmaBetter Verified Project**

Python, Pandas, Power BI, DAX | GitHub

- Analyzed 100,000+ customer records with Python and EDA to uncover credit-risk patterns across payment behavior, debt, credit utilization, and loan history, finding that 52% of customers were not paying the minimum amount due.
- Designed a 4-page Power BI dashboard with 8 custom DAX measures, converting the analysis into interactive KPIs and actionable risk insights for decision-making.
- Recommended repayment-reminder strategies to increase on-time payment rates among at-risk segments.

EDUCATION

M.S. in Computer Science: Machine Learning & AI Engineering, Woolf University — Jan 2025 – Present

Bachelor of Science, Mumbai University — May 2022

CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals — Microsoft, Aug 2026
- Applied Professional Certification in Data Science & Agentic AI — AlmaBetter, Aug 2026
- Applied Professional Certification in Data Analytics & Business Intelligence — AlmaBetter, May 2026